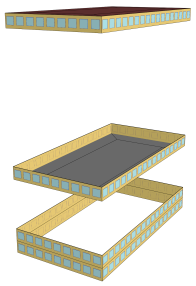
SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Office
Building subtype	High-rise (10+ storeys)
Vintage	Before 1983

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation.

BUILDING CHARACTERISTICS

Total floor area [m²]	31,532
Conditioned floor area [m²]	31,532
Above-ground storeys	8
Climate zone	7
Weather file	Montreal-Trudeau Intl AP QC CAN CWEC2016 WMO#=716270

WINDOW-TO-WALL RATIO [%]

Average	40.0
North	40.0
East	40.0
South	40.0
West	40.0

HVAC SYSTEM

Heating	Electric boiler (hot water) + Electric baseboards
Cooling	Water-cooled chiller
Air distribution	VAV with reheat (1 air loop)
Heat recovery	None

ENVELOPE & INTERNAL LOADS

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²K]	0.43
Roof [W/m²K]	0.22
Exposed floors / slabs [W/m²K]	2.94
Infiltration [m³/h/m² ext. surface]	0.82

GLAZING

Glazing U-factor [W/m²K]	2.50
SHGC	0.45

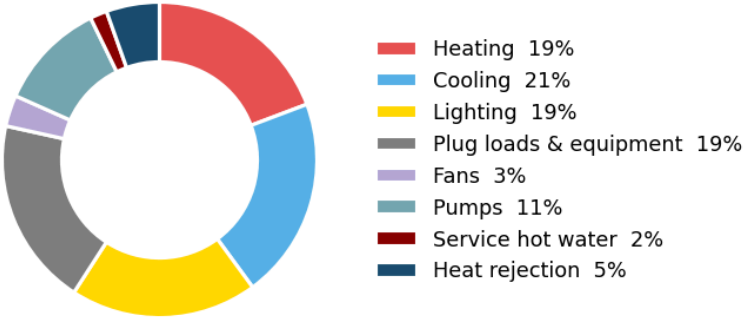
INTERNAL LOADS (MODEL INPUTS)

Occupant density [m²/person]	25.0
Lighting power density [W/m²]	8.8
Equipment power density [W/m²]	6.7

ANNUAL END USES [kWh/year]

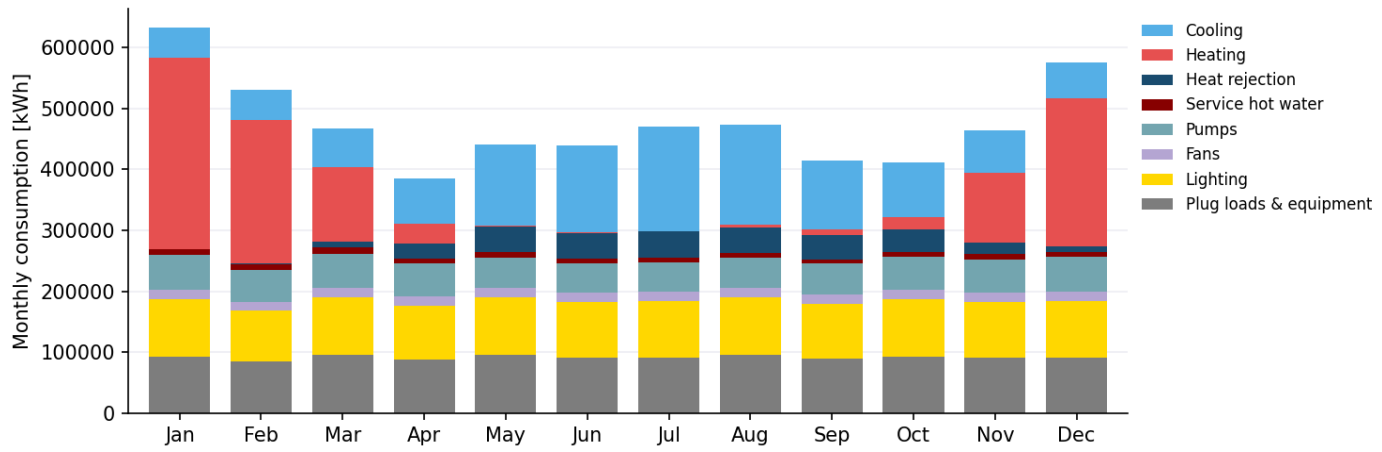
END USE	ELECTRICITY	TOTAL	kWh/m²
Heating	1,099,797	1,099,797	34.9
Cooling	1,179,533	1,179,533	37.4
Interior lighting	1,096,895	1,096,895	34.8
Plug loads & equipment	1,102,506	1,102,506	35.0
Fans	182,244	182,244	5.8
Pumps	635,611	635,611	20.2
Heat rejection	312,056	312,056	9.9
Service hot water	99,539	99,539	3.2
Total		**5,708,184**	**181.0**

SHARE OF ANNUAL CONSUMPTION



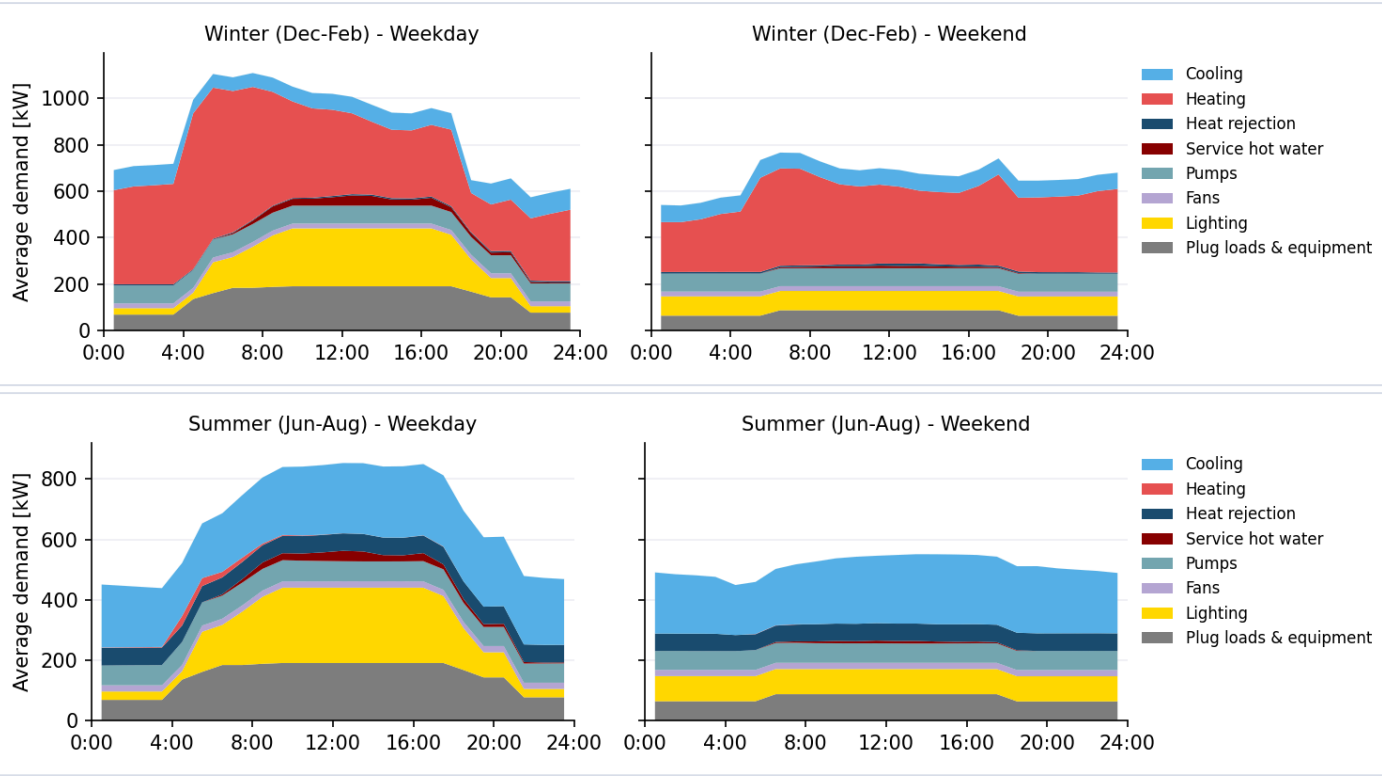
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS	PEAK ELECTRIC DEMAND
181 kWh/m²/yr (total area)	5,708,184 kWh/yr	5,708,181 kWh/yr	0 kWh/yr	2,086 kW · 66.1 W/m²

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption by end use, all fuels combined.

TYPICAL DAY PROFILES — WEEKDAY VS. WEEKEND



Average hourly demand by end use for winter and summer. Weekdays (Monday to Friday) are shown on the left and weekends and holidays on the right, showing the office occupancy schedule and HVAC set-back.

NOTES & ASSUMPTIONS

- Archetype energy model simulated with OpenStudio/EnergyPlus. Values are model outputs for the reference weather year and are not measured data.
- Energy use intensity is normalized by the total floor area (31,532 m²). Offices are not normalized per unit.
- Archetype: High-rise office (10+ storeys), built before 1983.
- HVAC entries marked "(typical)" are not modelled explicitly and come from the archetype assumption table.
- Peak electric demand occurs on 25-DEC-05:30.